

Omer Raza

MACHINE LEARNING ENGINEER

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Professional Summary

Machine Learning Engineer with 3+ years of experience deploying end-to-end ML systems in healthcare and logistics. Scaled CV, OCR, and risk models on AWS, via automation and CI/CD. Strong research foundation (EMNLP, IEEE) in GANs and LLMs; complemented by full-stack skills (React) to deliver production-ready solutions. Recognized with top-tier performance award.

Skills

Machine Learning: Computer Vision (Nerf, Gaussian Splatting, OCR, GAN, Yolo & OpenCV), LLM (Llama, Qwen), XgBoost, GNN
Deep Learning Frameworks: Pytorch & Hugging Face (extensive), Scikit-learn (extensive), TensorFlow (familiar), JAX (familiar)
Data Processing: Spark, SQL, Pandas, Numpy | **Cloud & MLOps:** AWS, Kafka, GitLab CI/CD, Docker
Programming Languages: Python, JavaScript, C++ | **Web & Mobile:** React & Electron

Industry Experience

Eagleview, USA - Multinational Photogrammetry & Aerial imagery Company Oct 2025 - present
Research Engineer III - Machine Learning

- **Project Lead** to carve insights, risks and bespoke implementations in **Gaussian Splatting** based models for improving visual fidelity and downstream metrics in existing processes such as mesh/DSM generation and pose optimization

Lalamove, Hong Kong - Multinational delivery logistics company (10M+ users, 100K+ daily transactions) Feb 2022 - May 2023
Machine Learning Engineer

Services & MLOps tools: Lambda API, EC2, Sagemaker, Rekognition, Real time/Batch processing jobs & Kafka.

- Led accuracy boost from **65% to 85%** in brand vehicle detection for market penetration tracking by standardizing label schema across teams, applying targeted augmentations (sticker masking) and model decoupling – saving **\$70K+** annually
- Developed and deployed object detection, OCR (with clustering heuristics) via automated evaluation and CI/CD (GitLab + AWS Lambda with Docker images), cutting driver bonus approval time from **2 days to 5 seconds**.
- Engineered a **risk assessment model (F-score: 0.95)** by implementing a full pipeline (data ingestion to ETL to feature engineering to real-time inference). This boosted daily fraudulent transaction detection by **20%**.

Received the highest performance award category in the year (4 people among 150 in the office)

Gense Technologies, Hong Kong - Award winning medical imaging (EIT) healthcare startup Nov 2020 - Dec 2021
Machine Learning & Software Engineer

- Deployed the **Gense Mobile App** (React) with live syncing to a proprietary EIT device, supporting user account and EIT test session management. This enabled rapid internal testing and real-time demos for partner investors.
- Developed iterative experimentation-driven ML models, achieving **88% EIT image quality** and **99% breathing alignment accuracy**, which enabled confidence-based diagnostics in the app.

Research Experience

RA Machine Learning at CIVS (Purdue University) on multimillion dollar grants for steel industry Sep 2023 – June 2025

- Architected a **silicon % prediction** pipeline for blast furnace (**90% acc**; beta testing at partner plant) to improve efficiency
- Developed a **hearth erosion model** (beta testing soon) by implementing first-principle models and inverse optimization techniques (3-5 min), later accelerated with **GNNs (20 sec)**, which can improve operator safety and campaign life.

RA Machine Learning (remote) at CUHK and HKUST Aug 2023 - Jul 2024

First/co-author of multiple ML papers on GANs & efficient LLMS in EMNLP & IEEE (Full paper list [here](#)). Sample paper below:

SR TGAN: Smoke Removal with Temporal GAN in robot assisted surgery

First Author – IEEE 2024

- Integrated temporal attention to improve smoke removal by **5–10%** over DeSmoke CycleGAN on JNBM/FADE metrics

Independent Research Project - Facial Expression Transformation via Latent Embedding Interpolation 2021

Automated **10K+** comic panel scraping and face extraction; explored GAN latent space (StyleGAN) for facial expression changes. Found that **Hyperplane-guided** vector shifts led to **stabler** generative outputs than direct transformation models.

Education

Masters in Computer Science at Purdue University, GPA 4.0/4.0 Aug 2023 - May 2025

Bachelors in Computer Science at The University of Hong Kong (HKU), First Class Honors Aug 2016 - Jun 2020

1) Deans Honor List, 2) HKU Foundation Scholar & 3) Young Tsun Dart Scholar (1 student / year / department)